PrimeWave Lab (Pvt) Ltd

TURNING COMPLEXITY INTO POSSIBILITY

SOFTWARE ENGINEERING INTERNSHIP

Technical Assessment

A practical full-stack development assessment designed to evaluate problem-solving, technical skills, and the ability to build reliable software solutions.

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Software Engineering Internship Technical Assessment

Assessment Title: Resource Reservation Management System
Assessment Type: Full-Stack Web Development
Submission Deadline: Thursday, 15 October 2026, 11:59 PM (Sri Lanka Time)

1. Introduction

At PrimeWave Lab, we believe that software engineering is more than simply writing code. It involves understanding real-world problems, identifying requirements, making technical decisions, and developing solutions that are practical, reliable, and maintainable.

This assessment is designed to evaluate your ability to approach a business problem and independently develop a working software solution.

You will be provided with a business scenario rather than a detailed technical specification. How you interpret the requirements, design the solution, and implement the application is entirely up to you.

2. Business Scenario

A growing organization manages several shared facilities and resources that are regularly used by different individuals and teams.

As the organization expands, coordinating resource usage has become increasingly difficult. Reservations are currently handled through informal communication and manual records, occasionally resulting in scheduling confusion, overlapping reservations, and uncertainty regarding resource availability.

The organization is looking for a centralized digital solution that simplifies how its resources and reservations are managed.

The proposed application should provide a practical way for the organization to organize its resources, coordinate reservations, and maintain a clear understanding of their usage and availability.

The organization expects the solution to remain reliable as its day-to-day activities grow.

3. Your Assignment

Your task is to design and develop a functional web application that addresses the business scenario described above.

You are responsible for interpreting the problem, identifying the necessary functionality, designing an appropriate data structure, and implementing a solution that you believe meets the organization's requirements.

There is no predefined application design or implementation approach.
We are interested in understanding how you independently translate a business requirement into a working software product.

Your solution should demonstrate thoughtful engineering decisions, practical usability, and appropriate consideration of real-world operational challenges.

4. Technical Expectations

The application should be developed using a modern full-stack JavaScript/TypeScript development approach.
Technology Stack

- Full-Stack Framework: Next.js (App Router)
- Language: TypeScript
- Database: PostgreSQL
- Version Control: Git and GitHub

You may select additional frameworks, libraries, and development tools based on your preferred approach.The application should use persistent data storage and be capable of running in a local development environment.

5. Development Approach

- You are free to determine the architecture, application structure, user experience, and development methodology.
- We encourage you to approach the assignment as you would a real-world software engineering task.
- Modern AI-assisted development tools, including OpenAI Codex, GitHub Copilot, and similar assistants, may be used to support your development process.
- Our evaluation will focus on your ability to produce a functional, understandable, and well-reasoned solution rather than the particular tools used during development.
- You should be prepared to explain your implementation and the decisions behind it.

6. Scope

- This assessment is intended to be a focused software engineering exercise rather than a large-scale enterprise application.
- You are not expected to build an extensive platform or introduce unrelated functionality.
- Focus on delivering a practical solution to the business problem within the available timeframe.
- We value thoughtful prioritization, reliability, and maintainability over unnecessary complexity.

7. Submission Requirements

1. GitHub Repository
Your submission must include a GitHub repository containing the complete application source code. The repository should include a README file with sufficient information for another developer to understand, set up, and run your solution.

2. Demonstration Video
A 3–5 minute screen recording demonstrating your completed application is also required. Your demonstration should communicate the purpose of your solution, how it addresses the business scenario, and the important technical decisions you made.

3. Entity Relationship (ER) Diagram
Provide an Entity Relationship (ER) diagram illustrating the database design of your application, including the relevant entities and their relationships. The diagram must be accessible through a shareable link submitted via the official assessment submission form.

8. Evaluation

Your submission will be reviewed based on the following areas:

- Understanding and interpretation of the business problem
- Problem-solving and engineering judgement
- Functional completeness and reliability
- Application architecture and database design
- Code quality and maintainability
- User experience and practical usability
- Technical understanding and communication

Different approaches are welcome, provided they demonstrate sound reasoning and address the underlying business requirements.

9. Submission Instructions

All completed assessments must be submitted through the official submission form:

Assessment Submission Form:
https://forms.gle/HzQH6tB5c4edPjnn8

- Submission Deadline: Thursday, 15 October 2026, 11:59 PM (Sri Lanka Time, UTC+05:30)
- The submission form will close at midnight, at the beginning of Friday, 16 October 2026.
- Submissions will not be accepted after the deadline.
- Please ensure that all required materials are accessible to our review team and submitted before the deadline.
- When completing the submission form, please use the exact email address you used to send your CV during your initial internship application.